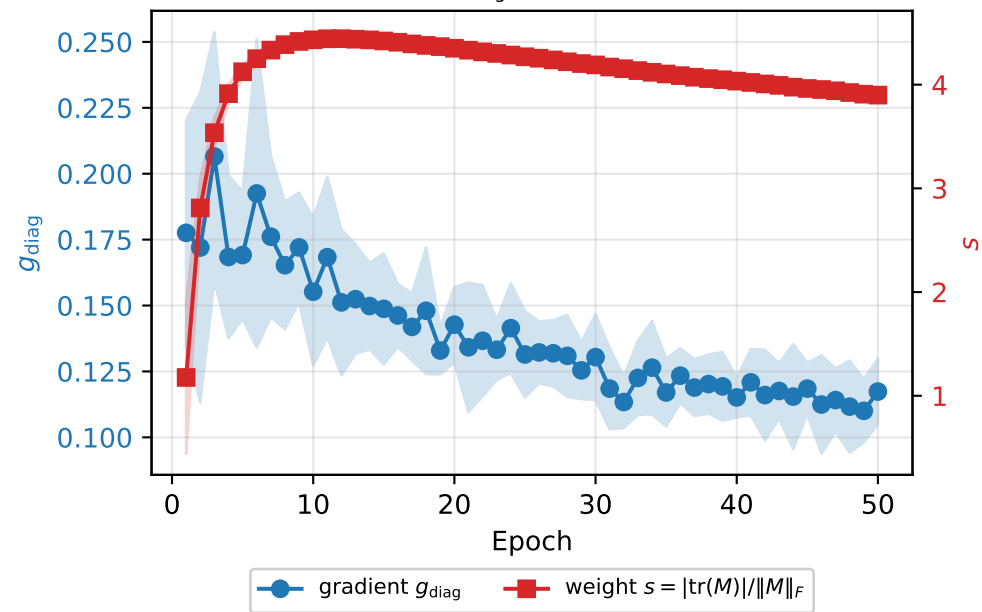
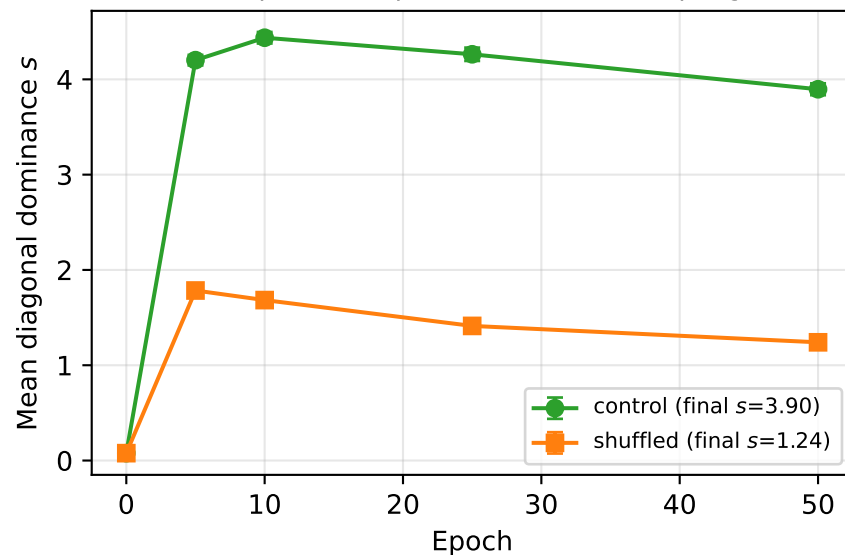


Gradient coupling mechanism (RQ1 §9): correlated $\nabla W_{in}, \nabla W_{out}$ updates build the diagonal fingerprint

A. Gradients stay flat while weights climb
Pearson $r(g_{\text{diag}}, s) = -0.12 \pm 0.09$



B. Shuffling ∇W_{out} cuts fingerprint by 68%
Independent updates break the coupling



C. Diagonal injection (no backprop) builds the fingerprint
Diagonal structure is sufficient

